HDOC DAY 48

PROBLEM

Write a C program that reads the size and elements of a variable-length integer array, arr[]. Pass the array to a function named pivot_index() using its address. Use pointers to pass the array between the calling function and pivot_index(). The function pivot_index() must:

- Calculate the left sum (sum of all elements strictly to the left of each index).
- Calculate the right sum (sum of all elements strictly to the right of each index).
- Treat the left sum = 0 for the first element.
- Treat the right sum = 0 for the last element.
- Find and display the leftmost index at which left sum = right sum.
- Display -1 if no pivot index exists.

CODE

```
#include <stdio.h>


int pivot_index(int *arr, int n)

{

  int total = 0, left_sum = 0, right_sum;


  for (int i = 0; i < n; i++)

  {

    total += *(arr + i);

  }


  for (int i = 0; i < n; i++)

  {

    right_sum = total - left_sum - *(arr + i);
```

```
        if (left_sum == right_sum)
        {
            return i;
        }

        left_sum += *(arr + i);
    }

    return -1;
}

int main()
{
    int n;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter the elements of the array: ");
    for (int i = 0; i < n; i++)
    {
        scanf("%d", arr + i);
    }

    printf("Pivot index = %d\n", pivot_index(arr, n));
```

```
    return 0;
}
```